

Opportunities in AR/VR/XR Productivity

Market Intelligence Report — 2025/2026

1. Overview & Scope

1.1 What is AR/VR/XR Productivity?

- **Extended Reality (XR)** is the collective term encompassing **Augmented Reality (AR)**, **Virtual Reality (VR)**, and **Mixed Reality (MR)** — technologies that blend or replace the physical world with digital content
- **XR Productivity** refers to the use of immersive technologies to enhance workplace output — including spatial desktops, 3D collaboration, AR-guided field work, immersive design reviews, and virtual co-presence
- **Core modalities:**
 - **AR** Digital overlays on the real world (field work, remote assistance, SOPs)
 - **VR** Fully immersive environments (virtual desktops, design, training)
 - **MR** Blended interaction with real + virtual objects (spatial workspaces, 3D review)

1.2 Why Now?

Driver	Detail
Hardware Maturity	Standalone headsets are more powerful, affordable, and comfortable; form factors shrinking toward everyday glasses
5G + Edge Computing	Low-latency connectivity and powerful edge processing enable real-time spatial experiences anywhere
AI Integration	49% of new AR/VR tools now feature AI-driven tracking, real-time behavior prediction, 3D object recognition, and automated content generation
Open Standards	OpenXR supported by 13+ major vendors — one codebase can target multiple devices
Enterprise ROI Proof	Market pivoting from gaming-centric to enterprise productivity; verifiable ROI in training, maintenance, and collaboration
Shifting Work Paradigms	The global shift to remote work exposed limitations of existing digital tools — video conferencing created "Zoom fatigue" and collaboration became a tedious game of screen sharing [2]

2. Market Size & Growth

2.1 Global XR Market Size (2024–2030)

Source	2024/2025 Value	2029/2030 Projection	CAGR
Knowledge Sourcing	\$33.9B (2025)	\$133.8B (2030)	31.6%
MarketsandMarkets	\$24.4B (2024) → \$37.9B (2025)	\$84.9B (2029)	28.3%
ABI Research (SW/Services)	\$44.7B (2024)	\$299.3B (2030)	~37%
Mordor Intelligence (Spatial Computing)	\$20.4B (2025)	\$85.6B (2030)	33.2%

- **Consensus:** Broad XR market sits in the **\$25B–\$45B range (2024–2025)**, projected to reach **\$85B–\$300B+ by 2030** depending on scope
- XR device shipments (e.g., lightweight AR or smart glasses for industrial use) grew over 40% year-over-year in 2025 [50]

2.2 CAGR & Growth Projections

- **Overall XR CAGR range:** 28%–53% across analyst reports (2024–2030)
- With over 1,380 companies working in this market and 145K+ employees, the domain continues to evolve at a steady pace, showing an annual growth rate of 27.92% [47]
- **Enterprise segment** growing faster than consumer, driven by industrial applications in remote collaboration, maintenance, and product visualization

2.3 Enterprise Adoption Rates & Key Data

- 52%+ of enterprise organizations report using AR/VR tools for operations, training, or customer engagement
- 56% of businesses have implemented some form of mobile AR/VR; another 35% are considering it
- Metrigy's research indicates "slow but steady" growth; a late 2024 survey of ~400 companies showed that 16.5% had targeted investment in XR technologies by the conclusion of 2025. [13]
- 29% of surveyed enterprises reported a 25%+ productivity increase; 61% say AR/VR resulted in 20%+ cost savings
- XR methods can reduce training time by 50% and cut operational costs while improving workforce readiness and productivity [41]

3. Key Products & Platforms

3.1 Spatial Computing Workspaces

Platform	Key Features	Status
Immersed	Up to 5 virtual monitors in Pro mode, giving users ample screen real estate to manage various tasks simultaneously. [5] Built the #1 AR/VR app with 1.5M users. Developing dedicated "Visor" hardware [4]	Active, growing
Horizon Workrooms (Meta)	VR meeting rooms with avatar presence, whiteboarding, integrated with Meta Quest ecosystem	Active
Microsoft Mesh Teams Immersive	On December 1, 2025, Microsoft retired the standalone Mesh app; experiences replaced by immersive events in Teams for interactive events that captivate attendees [12]	Pivoted
Spatial	Originally standalone 3D workspace; Shifted to a cross-platform 3D interactive experience hub for creators, enterprises, and educators across Web, Mobile, and VR	Active

Key Takeaway on Microsoft Mesh: Widespread demand for Mesh failed to ignite, and Microsoft retired it as a standalone app. [13] Team users can now host immersive events such as training sessions, virtual exhibitions, and product showcases. However, this is a significant downgrade — Microsoft Mesh supported up to 330 participants, whereas Teams Immersive Events supported only up to 16 participants, making it unsuitable for truly large gatherings. [13]

Immersed Deep Dive:

- Offers up to 5 virtual screens for Windows, Mac, or Linux computers, with support for resolutions up to 2560x1400 (Pro), including curved and portrait screens, and cross-platform compatibility [8]
- Developing the work-focused "Visor," which is claimed to be higher resolution than Apple Vision Pro, 70% lighter weight, and 70% less expensive [4]
- Combines virtual workspace, a work-focused headset, and an AI assistant that handles busywork [4]

3.2 Vision Pro Productivity Apps

Apple Vision Pro is described as an ideal productivity tool; its 3D user interface frees apps from the boundaries of a display, providing an infinite canvas for multitasking and collaboration. [22]

App / Suite	Highlights
Apple Native (Keynote, Pages, Freeform)	Freeform enables spatial brainstorming on an infinite board; Keynote offers a virtual conference room and stage environment for presentations [29]
Microsoft 365	Native Vision Pro versions of Office and Teams — draft documents or crunch spreadsheets on a massive virtual screen [29]
Fantastical	Makes full use of the infinite canvas and sharp text rendering to make daily tasks effortless [22]
JigSpace	Combines 3D content, audio, video, and text in interactive, step-by-step spatial presentations [22]
Craft, MindNode, Numerics	Spatial note-taking, mind mapping, and live KPI dashboards
Slack, Notion, Todoist	Thousands of popular productivity apps are compatible with Vision Pro at launch [22]

Limitations: *The infinite canvas is incredible technology, but the app ecosystem is still limited compared to iPad or Mac, and the \$3,499 price point is a significant barrier. [24]*

3.3 AR-Assisted Workflow Tools

Category	Key Players	Capabilities
Remote Expert Assistance	TeamViewer Assist AR, PTC Vuforia (Chalk → Zinc), Scope AR	AR allows experts to remotely guide on-site workers, eliminating physical travel; platforms like VIRNECT and TeamViewer enable experts to see what on-site workers see and provide guided steps through digital overlay [34]
AR Documentation & SOPs	Vuforia Expert Capture, Taqtile Manifest	Enables service providers to capture knowledge, scale work instructions, and complete AI-enhanced inspections [39]
AR-Guided Maintenance	Vuforia Studio, Dynamics 365 Guides (legacy)	Augmented work instructions and training that elevate skills, performance, uptime, safety, and overall efficiency [32]

ROI Note: *Most organisations achieve ROI within 6–12 months through reduced travel costs and faster resolution times, with some reporting 40% savings in dispatch expenses alone. [31]*

Vuforia Chalk Sunset: Due to PTC's acquisition of ServiceMax, Vuforia Chalk was officially retired on October 31, 2025, and its features were integrated into ServiceMax Zinc. [33]

9D Technologies | AI Research



3.4 Emerging / Notable Startups

Startup	Focus	Notable
Immersed	VR workspace + Visor hardware + AI assistant	1.5M users; pursuing NASDAQ listing
Taqtile (Manifest)	AR work instructions for manufacturing	Used by PBC Linear to convert expert knowledge into accessible digital guides, reducing training time significantly [50]
XREAL	Consumer/enterprise AR glasses	Invested \$55.3M+; partnered with BMW and Qualcomm to integrate AR into automotive and enterprise solutions [47]
Scope AR	AR remote assistance + work instructions	Enterprise-focused, multi-platform
JigSpace	3D spatial presentations	Native visionOS app for interactive walkthroughs

4. Use Cases & Workflows

4.1 Infinite Desktop / Multi-Monitor Replacement

- The primary superpower is virtual monitor creation — users can spawn multiple high-resolution screens of immense size, arranging them in curved wraparound format; the digital workspace is limited only by needs, not by wallet or desk size [\[7\]](#)
- With physical monitors, you're locked into whatever height and angle the stand allows; with a virtual screen, you can recline, shift posture — and the screen stays exactly where you need it [\[6\]](#)

4.2 3D Design & Review (CAD, Architecture, Product Design)

- Engineering, construction, and product development teams use XR to interact with digital twins, CAD models, and spatial simulations in shared environments [\[49\]](#)
- Devices like HTC VIVE Pro 2 enable high-fidelity visualization prior to physical production, reducing the need for iterative prototyping [\[49\]](#)

4.3 Remote Collaboration & Co-Presence

- AI acts as the invisible orchestrator of immersive workspaces — context-aware assistants can transcribe meetings in real-time, translate conversations, and pull up relevant documents when mentioned [\[2\]](#)
- Accenture's onboarding program at "One Accenture Park" has successfully integrated over 300,000 new hires through a virtual campus with monorail rides, educational exhibits, and gathering spaces [\[14\]](#)

4.4 AR Field Workflows (Maintenance, Logistics, Inspections)

- AR overlays work instructions utilizing smart glasses, making it easier for workers to understand and complete assembly processes [\[50\]](#)
- AR remote assistance is accelerating because it directly addresses skilled labour shortages; "see-what-I-see" models allow frontline technicians to receive contextual overlays and live expert guidance while preserving hands-free operation [\[49\]](#)

4.5 Data Visualization in 3D Space

- Numerics helps enterprises keep tabs on their business with live widgets that support more than 4,000 KPIs and can be placed anywhere in a user's space [\[22\]](#)
- Financial analysts can track vast arrays of live data across virtual screens; 3D charts and models enable pattern recognition impossible on flat displays

5. Enterprise Adoption Data

5.1 Companies Using XR for Productivity — Case Studies

Company	Use Case	Impact
Accenture	VR onboarding campus "One Accenture Park" — integrated over 300,000 new hires [14]	Scaled immersive onboarding globally
ConocoPhillips	Digital twins reducing facility staffing needs; 15% reduction in time on basic work orders; 90% decrease in time for preventive maintenance checks [41]	Massive maintenance efficiency
Krones AG	Uses TeamViewer xAssist for AR remote assistance; ~200 employees use it as a daily tool for remote customer support [34]	Reduced travel, faster resolution
bp	Built "HIVEs" (Highly Immersive Visualization Environments) to connect global teams; experts collaborate on wind turbines and equipment repairs via digital twins	Lower-cost, lower-carbon collaboration
Boeing / Tyson Foods	XR methods reducing training time by 50% and cutting operational costs; reported stellar ROI numbers [41]	Training and workforce readiness
DHL	Smart glasses for warehouse picking and logistics	Double-digit productivity improvements reported

5.2 ROI Data & Efficiency Gains

Metric	Data Point
Training Time	Up to 50% reduction via XR-based methods
Travel/Dispatch Costs	Up to 40% savings with AR remote assistance
ROI Timeline	6–12 months for AR remote assistance deployments
Maintenance Efficiency	15%–90% time reduction depending on task complexity (ConocoPhillips)
Cost Savings	61% of enterprises report 20%+ cost savings from AR/VR
Productivity	29% of enterprises reported 25%+ productivity increase

5.3 Barriers to Adoption

Barrier	Detail
Cost	Premium hardware (\$3,499 for Vision Pro; enterprise smart glasses \$1,500–\$4,000/device); bespoke content creation expensive
Comfort	Up to 40% of users report discomfort in full-immersion VR sessions; session duration limits needed
IT Integration	Enterprise AR platforms must balance cross-platform compatibility, secure data handling, and integration with existing IT service management systems [31]
Content Creation	Producing high-quality 3D content remains expensive and time-consuming
Cultural Resistance	Organizations need new norms and etiquette for the immersive workplace — training and change management are as important as the technology itself [2]
Limited Demand for General Meetings	Use cases tend to be "very targeted around training, product demonstrations, engineering and design, and customer engagement rather than for general purpose meetings [13]

6. Competitive Landscape Matrix

Product	Type	Platform	Key Features	Pricing (est.)	Traction
Immersed	VR Workspace	Quest, Vision Pro, PCVR	5 virtual screens, co-working, AI assistant	Free / Pro subscription	1.5M users
Horizon Workrooms	VR Meetings	Meta Quest	Avatar-based meetings, whiteboard, spatial audio	Free (with Quest)	Bundled with Meta ecosystem
Teams Immersive	3D Events	PC, Mac, Quest	Custom 3D events, branded environments, Unity toolkit	Teams Premium license	Enterprise (Accenture, bp)
Apple Vision Pro	Spatial OS	visionOS	Infinite canvas, native + 3rd party apps, Mac Virtual Display	\$3,499 (hardware)	Early adopter / premium
TeamViewer Assist AR	AR Remote Assist	Mobile, tablets, smart glasses	Live video + AR annotations, cross-platform	~\$50-100/user/month	Krones AG, 200+ daily users
Vuforia (PTC)	AR Work Instructions	Mobile, smart glasses, HoloLens	Expert Capture, Studio, AI inspections	Enterprise license	#1 Quadrant rank, 1M+ developers
Taqtile Manifest	AR Work Instructions	HoloLens, mobile	Step-by-step AR guides, knowledge capture	Enterprise license	PBC Linear, manufacturing focus
XREAL	AR Glasses	Proprietary	Lightweight consumer/enterprise AR glasses	~\$400-700	BMW & Qualcomm partnerships

7. Gaps & Whitespace

7.1 What's Missing in Current Solutions?

Gap	Detail
No Unified Cross-XR Workspace	A colleague on VR must be able to collaborate with someone on AR glasses and another on a laptop — cloud-based cross-reality collaboration is critical for widespread adoption but largely unsolved [2]
Comfort for All-Day Use	Even leading headsets fatigue users within 2-3 hours; true monitor replacement requires 8+ hour comfort
Text Readability	VR resolution still lags behind 4K monitors for extended reading/coding work
Seamless Input	The spatial interface creates new possibilities but also introduces friction that doesn't exist on conventional screens [24]
AI-Native Productivity	Most XR apps are spatial ports of 2D tools, not AI-first spatial applications
Content Creation Bottleneck	Creating 3D AR work instructions and environments remains costly and requires specialized skills
Standardized ROI Measurement	No industry-standard framework for quantifying XR productivity gains
SME Accessibility	High capital expenditure disproportionately affects small/medium enterprises

7.2 Where Could a Spatial AI Assistant Add Value?

Opportunity	Description
Context-Aware Workspace AI	An AI assistant that is context-aware — automatically transcribing meetings, translating conversations, pulling up relevant documents when mentioned [2]
Automated Task Distillation	AI that distills everything observed and interacted with into actionable tasks, insights, and reminders [4]
Intelligent Spatial Layout	AI that learns work patterns and dynamically arranges windows, screens, and tools optimally
LLM-Powered Field Guidance	Research prototypes have demonstrated LLM-powered XR assistants that provide context-aware guidance on assembly and maintenance tasks, reducing cognitive load on frontline workers [50]
Automated 3D Content Generation	AI that converts 2D documentation into interactive 3D spatial instructions, solving the content bottleneck
Meeting Intelligence	Real-time spatial transcription, action item extraction, and follow-up automation in virtual meeting rooms

8. Key Takeaways

1. **The market is real and growing fast** — XR productivity is projected at **28–37%** CAGR through 2030, with enterprise adoption leading the way
2. **Enterprise > Consumer** — There's less stylistic resistance in professional settings and a clearer business case for enterprise XR, such as productivity gains from line-of-sight guidance and greater efficacy in immersive training [\[42\]](#)
3. **AR remote assistance is the proven use case** — AR remote assistance improves first-time fix rates, reduces equipment downtime, and shortens escalation cycles; adoption grows because it removes friction from frontline work [\[49\]](#)
4. **Virtual desktops are maturing, not mature** — Immersed leads with 1.5M users, but comfort, resolution, and input friction still prevent mass adoption as a full monitor replacement
5. **Microsoft retreated; Apple entered premium** — Microsoft retired its Mesh 3D collaboration platform as it scales back metaverse ambitions, while Apple's Vision Pro targets high-end spatial computing [\[13\]](#)
6. **The AI + Spatial intersection is wide open** — Most current XR tools are spatial versions of 2D apps; an AI-native spatial assistant that understands context, automates tasks, and orchestrates workflows across apps represents a significant whitespace opportunity
7. **ROI is proven but narrow** — Strong results in training (50% time reduction), maintenance (15–90% time savings), and field support (40% dispatch savings); general office productivity ROI still unproven at scale
8. **The biggest barrier isn't technology — it's adoption** — Enterprise XR succeeds when organisations align people, process, and technology from the outset; change management, comfort, and content creation remain the critical bottlenecks [\[49\]](#)